



ALLIANCE
UNIVERSITY
Private University established in Karnataka State by Act No.34 of year 2010
Recognized by the University Grants Commission (UGC), New Delhi

NAAC
GRADE A+
ACCREDITED UNIVERSITY

Alliance School of Advanced Computing Bengaluru

Department of CSE

E1CSA313

Design Project-I Diary

Project/Batch ID	DP1-24-28-CSE-GEN-L-01		
Name of the Mentor	Dr. Rashmi Mothkur		
Title of the Project	AI-Powered Smart Campus Resource Management System		
Domain (Area)	Artificial Intelligence / Full-Stack & Cloud Computing		
Name of the Leader	Likith S		
Details of the students	Name	Registration Number	Branch
	Likith S	2411021061437	CSE
	Vinodkumar	2411021061425	CSE
	Nikhil kumar V	2411021061427	CSE
	Kalyan kumar T	2411021061421	CSE

Sign. of the Mentor

Sign. of the Project Coordinator-Dept.

Signature of the Chief Coordinator-ASAC

Project Details

Title of the Project:

AI-Powered Smart Campus Resource Management System

Description:

The AI-Powered Smart Campus Resource Management System is a modern, web-based management platform built to eliminate inefficiencies in educational resource allocation and facility monitoring. Traditional campus management relies on disconnected spreadsheets and rigid software that force administrators to manually track room bookings and leave facilities running unmonitored during vacant hours. This project solves this challenge by delivering an intelligent, unified software solution: an intuitive web dashboard featuring an interactive 3D campus view designed for administrators to oversee facilities in real time, and an automated background service that predicts utility demand and schedules classroom allocations. By connecting real-time simulated sensor telemetry with machine learning demand forecasting, the platform alerts facility managers to uncoordinated water pumping, timetable booking clashes, and unoccupied conditioned lecture halls. This ensures that administrators can make proactive, data-driven decisions that optimize classroom utilization and reduce unnecessary electricity and water waste entirely through software.

Scope:

To engineer a tightly scoped, highly functional software platform focusing strictly on two aspects of campus management: real-time facility resource visibility and automated, predictive timetable and utility optimization.

Objectives:

1. **Centralized Visual Monitoring:** Design an interactive web dashboard with an interactive 3D campus view to monitor classroom occupancy, energy usage, and water levels in real time.
2. **Predictive Demand Forecasting:** Develop machine learning models in Python to predict 24-hour campus electricity and water consumption based on historical patterns and schedules.
3. **Automated Timetable Optimization:** Implement a constraint satisfaction scheduling algorithm that eliminates 100% of room booking clashes and optimizes classroom allocation.
4. **Energy-Saving Recommendations:** Formulate intelligent software recommendation logic that identifies empty conditioned classrooms and alerts facility managers to reduce utility waste.

Outcomes:

- Significantly reduces administrative overhead by replacing manual spreadsheets with an intuitive, unified web dashboard for real-time facility oversight.
- Eliminates room booking overlaps and scheduling conflicts across academic blocks, maximizing classroom availability and student convenience.
- Provides actionable energy-saving insights by detecting unoccupied lecture halls and recommending timely cooling and lighting setbacks.
- Establishes a scalable, open-standards software foundation that enables data-driven campus sustainability without requiring expensive proprietary building software.

Problem Statement:

The current digital infrastructure connecting campus facilities and classroom scheduling is hindered by manual spreadsheets and disconnected management tools. This results in frequent timetable conflicts, empty rooms remaining fully lit and cooled, uncoordinated water pumping, and high electricity bills during peak hours. Furthermore, administrators lack a unified platform that provides real-time visibility into classroom occupancy and predicts future utility demand.

Abstract:

Modern university campuses face a common operational challenge: facility managers are burdened by disconnected spreadsheets, and institutions face high electricity costs, uncoordinated water pumping, and manual scheduling clashes. To resolve these systemic inefficiencies, this project presents the AI-Powered Smart Campus Resource Management System, a streamlined web-based software platform engineered for the Alliance University campus. The system provides an interactive web dashboard with an interactive 3D campus view built using React and Three.js, allowing administrators to monitor classroom occupancy, utility consumption, and environmental conditions in real time. A predictive machine learning pipeline analyzes historical campus records and simulated sensor streams to forecast electrical and water demand 24 hours in advance. In addition, an automated constraint satisfaction algorithm optimizes timetable schedules, eliminates room booking conflicts, and identifies unused conditioned classrooms to recommend energy-saving setbacks. Benchmark tests confirm smooth browser rendering at 60 FPS, fast telemetry processing under 100 milliseconds, and an estimated 18% reduction in peak energy demand through software-guided scheduling recommendations.

Weekly Meeting Record

Week No: _____

Date: _____ to _____

Design Project Title: _____

Group Members:

1. Name: _____ Reg. No. _____ ☐ Present

2. Name: _____ Reg. No. _____ ☐ Present

3. Name: _____ Reg. No. _____ ☐ Present

4. Name: _____ Reg. No. _____ ☐ Present

Meeting Venue: _____ ☐ On Time

Documents Submitted:

☐ Status Report

☐ On Time

Progress:

Issues:

Assessment of Progress:

☐ Excellent

☐ Good

☐ Satisfactory

☐ Fair

☐ Poor

Notes / Concerns / Comments: _____

Signature of the Mentor

Zeroth Review Evaluation Form

Date of Review:

Reviewer Comments: *(Provide specific feedback on the Title, Scope, Objectives, Outcomes, Abstract & Literature Review, Tools and Technologies supported and Presentation)*

Recommendations for Improvement: *(List suggestions or required changes before the next phase/review.)*

Mentor Signature

Name:

Date:

Examiner 1 Signature

Name:

Date:

Examiner 2 Signature

Name:

Date:

First Review Evaluation Form

Date of Review:

Reviewer Comments: *(Provide specific feedback on the Problem Statement, Applications / Use Cases, Background Work, Feasibility Study, Methodology, Design, Technologies and Tools, Partial Implementation – 50% and Presentation)*

Recommendations for Improvement: *(List suggestions or required changes before the next phase/review.)*

Mentor Signature

Name:

Date:

Examiner 1 Signature

Name:

Date:

Examiner 2 Signature

Name:

Date:

Second Review Evaluation Form

Date of Review:

Reviewer Comments: *(Provide specific feedback on the Requirement Analysis, Module Description, Convergence, Coverage and Completeness towards the Idea, Implementation, Paper, Comprehensive Report and Presentation)*

Recommendations for Improvement: *(List suggestions or required changes before the Final Presentation in Semester End Examination.)*

Mentor Signature

Examiner 1 Signature

Examiner 2 Signature

Name:

Name:

Name:

Date:

Date:

Date:

Semester End Examination Evaluation

Date of Review:

Reviewer Comments: *(Provide specific feedback on the Project Report submitted, Problem Formulation, Review of Literature, Design Methodology, Implementation and Concept of the Work / Originality and Significance & Relevance – Application in the Real World.)*

Project Submission Checklist

Item	Yes	No	Remarks
Project Report Submitted	☐	☐	
Survey Paper Submitted in Conference	☐	☐	
Source Code / Software Submitted (if applicable)	☐	☐	
Prototype or Model Demonstrated (if applicable)	☐	☐	
Presentation Slides Submitted	☐	☐	
Project Diary Maintained	☐	☐	
Ethical Clearance Obtained (if applicable)	☐	☐	
Any Other Supporting Documents	☐	☐	

Examiner 1 Signature

Name:

Date:

Examiner 2 Signature

Name:

Date:

Student Consent Form

Academic Year: _____

Semester: _____

Degree Program: _____

Project Title: _____

Mentor Name: _____

Mentor's Designation: _____

Team Members

Sl. No.	Name of the Student	Register Number
1.		
2.		
3.		
4.		

Declaration by the Students

We hereby confirm that we have chosen the above topic for our Design Project – I as part of the curriculum for the _____ semester of our _____ degree program.

We understand and agree to the following:

1. The project work is academic in nature and must follow the guidelines.
2. We will carry out the project work ethically and responsibly under the supervision of the assigned Mentor.
3. We will not engage in plagiarism or submit any copied or previously submitted work.
4. We will attend all review meetings and submit documentation as per the department schedule.
5. Any form of academic misconduct or non-compliance may lead to disqualification or disciplinary action.

Signatures of Students

Sl. No.	Name of the Student	Signature
1.		
2.		
3.		
4.		

Design Project – General Rules

1. All projects must be approved by the supervising faculty before starting.
2. All work must be original; plagiarism is strictly prohibited.
3. Students can work in teams.
4. Teams must have 4 members.
5. Roles and contributions of each team member must be clearly defined.
6. A team leader must be designated for communication and coordination.
7. Team members must contribute fairly and equally.
8. Conflicts must be reported promptly to the Faculty In-charge / Coordinator.
9. Students must meet regularly with their Mentor and Faculty In-charge of the class weekly.
10. Meeting notes or progress reports must be maintained.
11. Follow official timelines for proposals, drafts, final reports, and presentations.
12. Maintain the documentation phase by phase.
13. Projects must follow ethical guidelines (no falsified data, respect IP, consent/privacy for surveys).
14. Follow the formatting guidelines (APA, IEEE, etc.) for all submissions.
15. Submit all required documents: proposal, report, logbook, and presentation materials.
16. Deliver a final presentation and be prepared for questions.
17. Maintain professional conduct with Mentor, Faculty In-charges, Teammates, and evaluators.
18. Use institutional resources responsibly and uphold academic integrity.